COL2010: Data Science

End-Semester Examination — Module 2: Data Cleaning and Preprocessing

Set D

Time: 3 Hours **Total Marks: 100**

Instructions to candidates.

- This paper contains **15** questions. Attempt **all** questions.
- All numerical answers must retain at least four significant figures unless an exact closed form is requested.
- All claims of the form "prove", "show", or "derive" require a fully rigorous argument; a correct final formula with no justification earns no credit.
- State any assumption you invoke that is not already given in the question.
- Calculators are permitted; symbolic computer algebra devices are not.

Q1. A university enrollment system compares student-name strings using the Levenshtein edit distance $\mathrm{lev}(a, b)$, defined as the minimum number of single-character insertions, deletions, and substitutions transforming string a into string b.

(a) Prove that lev is non-negative, satisfies $\mathrm{lev}(a, b) = 0 \iff a = b$, and is symmetric.

(b) Prove the triangle inequality $\mathrm{lev}(a, c) \leq \mathrm{lev}(a, b) + \mathrm{lev}(b, c)$ rigorously. (*Hint:* an edit script transforming a into b in $\mathrm{lev}(a, b)$ steps, followed by an edit script transforming b into c in $\mathrm{lev}(b, c)$ steps, is *some* valid sequence of edits transforming a into c; argue why this immediately bounds the *minimum*-length script from a to c.)

(c) The normalized similarity used in the course is $\mathrm{sim}(a, b) = 1 - \mathrm{lev}(a, b)/\max(|a|, |b|)$. Give an explicit example of three strings a, b, c for which sim (rather than lev itself) violates the triangle-inequality-analogue $d_{\mathrm{sim}}(a, c) \leq d_{\mathrm{sim}}(a, b) + d_{\mathrm{sim}}(b, c)$ where $d_{\mathrm{sim}} = 1 - \mathrm{sim}$, or prove that no such example exists.

[8 marks]

Q2. Final exam scores (out of 100) for $n = 9$ students in one section are:

$$20,\ 62,\ 65,\ 66,\ 68,\ 70,\ 71,\ 73,\ 75.$$

Compute Q_1, the median, Q_3, and IQR via the median-of-halves convention, and hence the Tukey fences. State which observation(s) are flagged, and note which *side* of the distribution the flagged value lies on. **[4 marks]**

Q3. A learning-management system suffers a submission-timeout bug: a fraction $\varepsilon \in (0, 1)$ of quiz submissions are logged with an identical placeholder score M (representing "ungraded"), while the remaining submissions have genuine scores confined to a bounded range negligible relative to M in the limit $M \to \infty$.

(a) Derive $\mu(M)$, $\sigma(M)$, and show the ordinary z-score of a placeholder entry converges to $\sqrt{(1 - \varepsilon)/\varepsilon}$ as $M \to \infty$.

(b) Solve for ε^* at which this limit equals 3, and state the operational conclusion: what fraction of a batch being logged identically is enough to make the classical 3σ rule permanently blind to the glitch, however extreme the placeholder value?

(c) Prove the modified z-score based on median/MAD instead diverges to $+\infty$ for any fixed $\varepsilon < 0.5$. A teaching assistant claims "mean/std breaks down the moment there is even one outlier, so it is categorically inferior to median/MAD." Using your results in (a)–(c), state precisely which notion of breakdown this claim conflates, and give the numeric threshold at which each notion actually fails for this scenario.

[9 marks]

Q4. Two students are described by (Attendance %, Accumulated Library Fine in paise): $P = (72, 4500)$, $Q = (95, 6200)$.

(a) Compute the raw Euclidean distance and identify the dominating coordinate.

(b) Given cohort statistics $\mu_{\text{att}} = 80, \sigma_{\text{att}} = 10$ and $\mu_{\text{fine}} = 5000, \sigma_{\text{fine}} = 900$, standardize both coordinates and recompute the Euclidean distance.

(c) A "similar academic risk profile" clustering tool is run on raw features. Explain, quoting your two distances, why recording the fine in paise rather than rupees would silently make attendance irrelevant to the clustering.

[6 marks]

Q5. Two students' enrolled-course tags over {CS101, MATH201, PHY101, CHEM101, BIO101} are $x = (1, 1, 0, 0, 1)$ and $y = (1, 0, 0, 1, 1)$.

(a) Compute the Jaccard similarity and cosine similarity of x and y.

(b) A third student's enrollment vector $z = (0, 0, 0, 0, 0)$ reflects an incomplete registration record, not a genuine claim of zero enrollments. State $\cos(z, y)$ and $\cos(z, z)$ under the course convention, and explain why silently treating z as a genuine "no courses" vector could corrupt a course-overlap-based advising tool.

[4 marks]

Q6. Let R denote the missingness indicator for a column $Y = (Y_{\text{obs}}, Y_{\text{mis}})$, with mechanism $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}; \phi)$.

(a) State Rubin's formal definitions of MCAR, MAR, and MNAR.

(b) Prove the ignorability result under MAR (with distinct parameters), showing the observed-data likelihood factors cleanly into data-model and mechanism terms.

(c) A university's self-reported "weekly study hours" field is disproportionately left blank by students who study the least, i.e. $P(R = 0 \mid \text{StudyHours})$ increases as true StudyHours decreases. Classify this mechanism formally, and use your proof in (b)–(c) to explain why no imputation using only observed fields can, even in principle, remove the resulting upward bias in average reported study hours.

(d) The university also has an always-observed LMS login-duration signal correlated with true engagement. Explain, in general terms, how conditioning on this variable could move the analysis toward a recoverable MAR-type situation relative to the enlarged observed set, and why this demonstrates that the MCAR/MAR/MNAR classification is relative to the observed variable set, not an intrinsic property of StudyHours alone.

[8 marks]

Q7. An analyst applies iterative predictive imputation to five student records over GPA, Attendance (%), and weekly Study Hours:

Row	GPA	Attendance	StudyHours
1	8.2	88	14
2	7.5	75	**?**
3	6.8	65	9
4	9.0	**?**	20
5	7.0	70	11

Missing cells are initialized with the column mean over observed values. The cyclic order is **StudyHours** $\rightarrow$ **Attendance**, using the fitted relations

$$\widehat{\text{StudyHours}} = 1.2{\cdot}\text{GPA}{+}0.05{\cdot}\text{Attendance}{+}1, \qquad \widehat{\text{Attendance}} = 5{\cdot}\text{StudyHours}{-}3{\cdot}\text{GPA}{+}40.$$

(a) State the two initial fill values.

(b) Perform one full iteration (Row 2's StudyHours, then Row 4's Attendance), showing all substituted numbers.

(c) Explain, referencing the specific predictor columns involved, why a second iteration leaves both values unchanged here.

(d) Your computed value for Row 4's Attendance exceeds 100%, which is outside the valid domain for a percentage. Explain what this reveals about the danger of applying a linear predictive-imputation model outside the numerical range on which its coefficients are meaningful, and state one concrete safeguard a practitioner should add to the pipeline.

[6 marks]

Q8. A university identity-resolution system linking Registrar and LMS records computes pairwise similarities for three student records A, B, C: $s(A,B) = 0.78$, $s(B,C) = 0.76$, $s(A,C) = 0.20$, with lower threshold 0.4 and upper threshold 0.7.

(a) Classify each pair.

(b) State the connected-components clustering of $\{A, B, C\}$ under "match" edges, and explain the transitivity problem.

(c) With $\text{Obj}(\mathcal{C}) = \sum_{i,j \text{ same cluster}}(s(i,j)-\theta)$, $\theta = 0.5$, compute Obj for $\{A, B, C\}$ versus $\{A, B\}, \{C\}$, and conclude which is objectively preferable.

(d) For a chain of n student records with adjacent similarity s_0 (just above upper threshold) and non-adjacent similarity floor f (below lower threshold), derive Obj exactly for (i) the single connected-component clustering and (ii) the clustering retaining only adjacent pairs, and prove the gap is $\Theta(n^2)$. State the operational lesson for a university merging student identities across several independent campus systems accumulated over many years.

[10 marks]

Q9. Accumulated library fines (in Rs.) for nine students are strongly right-skewed: 5, 8, 12, 18, 25, 45, 90, 22
Compute $\log_{10}$ of each to two decimal places, and state the raw ratio max / min against the transformed range, quantifying the compression achieved. **[4 marks]**

Q10. A university analytics team fits a scaler for *Age at Enrollment* and a one-hot encoder for *Major* on the full cohort (training and test rows combined) before splitting, to predict first-year dropout.

(a) With $n_{\text{tr}}, n_{\text{te}}$ the train/test sizes and $\mu_{\text{tr}}, \mu_{\text{te}}$ the true population Age means restricted to each subpopulation (which may differ if the test cohort is drawn from a later admission year), derive the exact bias of the leaked full-data mean relative to the correct train-only mean, and state the condition under which it vanishes.

(b) Explain formally why this bias implies any downstream test-set dropout-prediction metric is optimistically biased.

(c) A newly launched *Major* (e.g. "Data Science") appears only among test-year students. State precisely how a leak-free encoder fitted on training categories only must represent this row, and explain why fitting the encoder on the union of all majors seen across the whole dataset is still leakage.

[8 marks]

Q11. Two candidate duplicate student records are compared on four fields: name similarity = 0.92 (Jaro–Winkler), enrollment-number agreement = 1, phone agreement =`NaN` (missing on one record), date-of-birth agreement = 0. Weights: name 0.4, enrollment-number 0.3, phone 0.2, DOB 0.1.

(a) Compute the weighted-mean score using only available fields, weights renormalized.

(b) Classify against thresholds lower = 0.5, upper = 0.75.

(c) Explain why substituting 0 for the missing phone field instead of excluding and renormalizing would bias the decision, and in which direction.

[6 marks]

Q12. Self-reported weekly study hours for $n = 7$ students: 10, 11, 9, 12, 10, 11, 70.

(a) Compute the mean, standard deviation, and ordinary z-score of the value 70.

(b) Compute the median, MAD, and modified z-score of 70.

(c) Using $|z| \geq 3$ and $|z_{\text{mod}}| > 3.5$, state which method(s) flag 70, and connect the disagreement explicitly to your general result in Q3.

[4 marks]

Q13. Let X, Y be two numerical features (e.g. GPA and Attendance %) with means μ_X, μ_Y, standard deviations $\sigma_X, \sigma_Y > 0$, and Pearson correlation $r(X,Y) = \text{Cov}(X,Y)/(\sigma_X\sigma_Y)$. Let $X' = (X - \mu_X)/\sigma_X$ and $Y' = (Y - \mu_Y)/\sigma_Y$ be their z-score standardizations.

(a) Prove rigorously that $\text{Var}(X') = 1$ and $\text{Var}(Y') = 1$.

(b) Prove rigorously that $\text{Cov}(X', Y') = \text{Cov}(X,Y)/(\sigma_X\sigma_Y)$, and hence that $r(X', Y') = r(X,Y)$: z-score standardization exactly preserves Pearson correlation.

(c) State and justify the general lemma of which this is a special case: Pearson correlation is invariant under any strictly increasing *affine* transformation applied independently to each variable (i.e. $X \mapsto aX + b$, $Y \mapsto cY + d$ with $a, c > 0$).

(d) Using part (c), identify which transformations among {Min–Max scaling, robust (median/IQR) scaling, log transform, Box–Cox/Yeo–Johnson, discretization} are guaranteed to preserve Pearson correlation exactly, and which are not (even though a monotonic one among the latter may preserve *Spearman rank* correlation). Justify each classification in one sentence.

[8 marks]

Q14. Two blocking passes generate candidate duplicate student pairs among $n = 4000$ standardized records linking Registrar and LMS systems. Pass 1 (same declared major and birth year) yields blocks of sizes $45, 60, 30$. Pass 2 (same last-four registered-phone digits) yields blocks of sizes $8, 11, 6, 15$. The two candidate sets overlap in 40 pairs.

(a) Compute the candidate pairs from each pass and the size of their union.

(b) Compute the reduction ratio RR for the union relative to $\binom{n}{2}$.

(c) Ground truth contains 300 true duplicate pairs; Pass 1 alone captures 260, Pass 2 alone captures 150, both jointly capture 120. Verify via inclusion–exclusion the pair completeness of the union and state its numerical value.

(d) Prove $1-(1-p_1)(1-p_2) \geq \max(p_1, p_2)$ for independent capture probabilities $p_1, p_2 \in [0, 1]$, and relate it to your answer in (c).

[6 marks]

Q15. A university registrar gives you eight student records:

ID	Semester	Attendance (%)	Exam Score	Final GPA	Name (standardized)
1	4	88	72	8.1	ananya iyer
2	3	82	68	7.6	ananya iyer
3	6	85	70	**?**	rohan mehta
4	7	90	95	9.2	kabir singh
5	5	96	30	7.4	diya kapoor
6	8	80	65	**?**	s. thomas
7	6	86	71	7.9	rohan mehta
8	4	84	69	7.7	priya nambiar

Among the six rows with observed Final GPA, the mean Semester is 4.83; among the two with missing Final GPA, the mean Semester is 7. From the other seven rows (excluding Row 5), $\mu = (\mu_A, \mu_E) = (85.0, 70.0)$ for (Attendance, Exam Score) and $\Sigma = \begin{bmatrix} 16.0 & 2.0 \\ 2.0 & 81.0 \end{bmatrix}$. Rows 3 and 7 are suspected duplicate student records, with comparison vector: name similarity 0.95, ID-number agreement `NaN` (missing on one system), department agreement 1, DOB agreement 0, using the weights name 0.4, ID-number 0.3, department 0.2, DOB 0.1, and thresholds lower $= 0.5$, upper $= 0.75$.

(a) State, with the standard caveat about MNAR being unprovable from observed data alone, what the semester-conditioned evidence does and does not establish about the missingness mechanism of Final GPA.

(b) Compute, by hand, the Mahalanobis distance of Row 5's (Attendance, Exam Score) $= (96, 30)$ from μ, explicitly computing Σ^{-1}, and comment on whether Row 5 is a multivariate outlier despite neither coordinate looking extreme in isolation.

(c) Decide, using the weighted comparison score, whether Rows 3 and 7 should be merged, and construct the merged record's Final GPA field using the course's numeric golden-record rule.

(d) Argue formally whether deduplication should precede or follow imputation here, prove your ordering strictly dominates the reverse for at least one field, and state the audit-trail invariant that must survive regardless of ordering.

[9 marks]